

Day 23 Observation Guide: Equivalent Ratios with Mixed Numbers

OBSERVER GUIDE

Grade 6 Mathematics · Professor Xavier Honablué, M.Ed. · For visiting guests and observers

Purpose of This Guide

Days 21 and 22 built equivalent ratios with whole-number multipliers: equal groups, double number lines, and ratio tables. Day 23 removes that safety net. Ian rides his unicycle 10 yards every 4 seconds; how long does 25 yards take? No whole number takes 10 to 25. Students must either **go through a smaller ratio** ($10 : 4 \rightarrow 5 : 2 \rightarrow 25 : 10$) or scale once by the **mixed-number scale factor $2\frac{1}{2}$** . The lesson then carries that skill into real mixtures, where students **cross between part-to-part and part-to-whole ratios** and confirm equivalence with cross products.

This guide explains the design, maps today to the i-Ready lesson, and lists every material and research source used to build it.

The Design Decision: Two Roads, Then the Real World

We teach two strategies on purpose. Going through a smaller ratio uses only moves students already trust (divide both, multiply both) and always works. The single mixed-number multiplier is faster and connects ratio reasoning to fraction multiplication (5.NF.B.4, 6.NS.A.1). Students who hold both can check one against the other.

The real-life segment is where ratios usually break for sixth graders: a recipe is written part-to-part, but you buy and fill the *whole* — the pitcher, the wheelbarrow, the paycheck. Scaling a part-to-part ratio against the whole is the most common real-world error, so the app detects that specific wrong answer and names it.

Mixed Numbers Get Taught, Not Assumed

The featured error is Devon's: $4 \times 2\frac{1}{2} = 8\frac{1}{2}$. He multiplies the whole-number part and attaches the fraction. We answer it three ways: the distributive property ($4 \times 2 + 4 \times \frac{1}{2}$), a picture of $2\frac{1}{2}$ groups of 4 seconds, and a cross-product check showing $25 : 8\frac{1}{2}$ is not equivalent to $10 : 4$ ($100 \neq 85$).

i-Ready Classroom Mathematics Alignment

Today's lesson is drawn from i-Ready Classroom Mathematics, Grade 6, Unit 3, Lesson 13 ("Find Equivalent Ratios"), Session 4 — Develop: Using Equivalent Ratios, and previews Lesson 14 ("Use Part-to-Part and Part-to-Whole Ratios").

i-Ready content (Grade 6, Unit 3)	Where it appears today	Standard(s)
L13 S4 · Start: Same and Different (2 to 3, 12 : 6, 6 : 9, 18 to 9)	App step 2; worksheet Do Now	6.RP.A.1, MP.7
L13 S4 · Try It: Ian's unicycle, 10 yd every 4 s $\rightarrow$ 25 yd	App steps 3–5; worksheet Section C	6.RP.A.3a
L13 S4 · Apply It: the baker (96 : 72 $\rightarrow$ 36 : ?)	App step 6	6.RP.A.3a
L13 S4 · Practice Example: shampoo (32 fl oz : \$8 $\rightarrow$ 12 fl oz)	App step 5; worksheet Section C	6.RP.A.3a, 6.RP.A.3b
L13 S4 · Close: Exit Ticket — 56-floor skyscraper, 45 m per 10 floors	App step 9	6.RP.A.3
Preview of L14 · part-to-part vs part-to-whole	App step 7; worksheet Section D; anchor chart	6.RP.A.1, 6.RP.A.3

Teacher note: session numbers follow the i-Ready Classroom Mathematics Grade 6 materials. Confirm exact page numbers against your edition before sharing this guide with observers.

Lesson Flow — 55 Minutes

Min	Segment	What the class is doing	Look for
0-3	Welcome back	Observer note from Day 22; learning targets.	Students predicting whether 25 yd takes more or less than 8 s.
3-8	Same and Different	Four ratio cards; test any pair in the app.	Someone noticing that $B \rightarrow D$ is $\times 1\frac{1}{2}$.
8-14	Ian's Unicycle	Three Reads; solve with any tool.	The stall at “ 10×2 is too small, 10×3 is too big.”
14-20	Go through a smaller ratio	$10 : 4 \rightarrow 5 : 2 \rightarrow 25 : 10$ on a double number line and table.	Students choosing a divisor that makes the target reachable.
20-27	Mixed-number multiplier	$25 \div 10 = 2\frac{1}{2}$; trail mix ($2\frac{3}{4} : 1\frac{1}{2}$); shampoo ($\times \frac{3}{8}$).	Predicting bigger or smaller before multiplying.
27-33	Devon's error + the baker	Error analysis; Apply It multiple choice.	“Both parts of the mixed number multiply.”
33-41	Real life: part $\leftrightarrow$ whole	Lemonade, concrete, paint, savings, free throws; cross-product checker; class-data question.	Students adding the parts <i>before</i> scaling to a total.
41-49	Recipe Lab	Teams scale a smoothie recipe to an order; submissions are checked live.	A written scale factor on every submission.
49-55	Stations + exit ticket	Engage / Explore / Enrich; skyscraper exit ticket.	$5\frac{3}{5}$ or a smaller-ratio path in the written work.

Predictable Student Errors — and What We Do About Them

Error	Why it happens	Our move today
$4 \times 2\frac{1}{2} = 8\frac{1}{2}$	Multiplying only the whole-number part of a mixed number.	Distributive property, $2\frac{1}{2}$ -group picture, cross-product check.
Baker: $132 (96 + 36)$ or $12 (36 - 24)$	Additive thinking — keeping the difference instead of the ratio.	Name it as the same error from Days 20-22; ratios are multiplicative.
Lemonade: 10-cup pitcher $\rightarrow 3\frac{3}{5}$ cups concentrate	Scaling the part-to-part ratio ($1\frac{1}{2} : 4\frac{1}{2}$) against the whole.	“Cross to the whole first”: $1\frac{1}{2} + 4\frac{1}{2} = 6$, then $10 \div 6$.
Calling $12 : 15$ (girls : boys) part-to-whole	Not checking what the second number counts.	Sentence frame: “The ___ is compared to the ___.”
Scale factor written upside down ($10 \div 25$)	Dividing original $\div$ new.	Predict first: should the answer get bigger or smaller?

Real-Life Connections Used Today

Context	Part : part	Part : whole	Scaled to	Answer
Lemonade stand (food service)	$1\frac{1}{2} : 4\frac{1}{2}$	$1\frac{1}{2} : 6$	10-cup pitcher ($\times 1\frac{2}{3}$)	$2\frac{1}{2}$ c concentrate, $7\frac{1}{2}$ c water
Mixing concrete (construction)	$1 : 2\frac{1}{2}$	$1 : 3\frac{1}{2}$	14 shovels ($\times 4$)	4 cement, 10 sand
Mixing paint (art, auto body)	$2\frac{1}{4} : \frac{3}{4}$	$2\frac{1}{4} : 3$	5 quarts ($\times 1\frac{2}{3}$)	$3\frac{3}{4}$ qt blue, $1\frac{1}{4}$ qt yellow
Save & spend (finance)	$2\frac{1}{2} : 10$	$2\frac{1}{2} : 12\frac{1}{2}$	$\$87\frac{1}{2} (\times 7)$	$\$17\frac{1}{2}$ saved, $\$70$ spent
Free throws (sports analytics)	$3 : 2$	$3 : 5$	40 attempts ($\times 8$)	24 made, 16 missed
Class data (cross-product check)	$12 : 15$ and $8 : 10$	$12 : 27$ and $8 : 18$	—	Both $4 : 9$ ($12 \times 18 = 216 = 27 \times 8$)

Answer Key

App: Ian 25 yd $\rightarrow$ 10 s · Trail mix 9 c peanuts $\rightarrow$ 6 c raisins · Shampoo 12 fl oz $\rightarrow$ \$3 · Baker $\rightarrow$ 27 banana (B) · Recipe Lab: Order 1 $\times$ 3 $\rightarrow$ $4\frac{1}{2}$: $2\frac{1}{4}$; Order 2 $\times$ 3 $\rightarrow$ $4\frac{1}{2}$: $2\frac{1}{4}$; Order 3 $\times$ $2\frac{1}{2}$ $\rightarrow$ $3\frac{3}{4}$: $1\frac{7}{8}$; Order 4 $\times$ $1\frac{1}{2}$ $\rightarrow$ $2\frac{1}{4}$: $1\frac{1}{8}$; Order 5 $\times$ $\frac{1}{2}$ $\rightarrow$ $\frac{3}{4}$: $\frac{3}{8}$ · Practice: 120 mi; $6\frac{2}{3}$ c; $1\frac{1}{2}$ qt · Exit ticket: $56 \div 10 = 5\frac{3}{5}$, $45 \times 5\frac{3}{5} = 252$ m.

Worksheet: Do Now: A & C, B & D; B $\rightarrow$ D is $\times 1\frac{1}{2}$ · A: SCALE FACTOR; $25 \div 10 = 2\frac{1}{2}$ · B: 1 part-to-part, 2 part-to-whole, 3 mixed number, 4 cross products · C: $\frac{3}{8}$; \$3 · D: lemonade $1\frac{1}{2}$: 6, $2\frac{1}{2}$ c; concrete 1 : $3\frac{1}{2}$, 4 shovels; savings $2\frac{1}{2}$: $12\frac{1}{2}$, \$17 $\frac{1}{2}$; free throws 3 : 5, 24 made; cross check $216 = 216$, yes.

Materials

- Day 23 interactive lesson app — cpamath6day23.streamlit.app (projected; students follow along).
- Day 23 worksheet (ELA + Math word study) — [Day23_wksht.pdf](#).
- Anchor chart: Part-to-Part & Part-to-Whole — [Anchor_Chart_Part_to_Part_and_Part_to_Whole.pdf](#).
- i-Ready Classroom Mathematics, Grade 6, Lesson 13 Session 4 slides and Student Book pp. 297–302.
- Fraction strips, double-number-line strips, measuring cups ($\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$, 1 cup), grid paper.
- Homework: Khan Academy — Equivalent Ratio Word Problems (90% target), all work shown on paper.

Research Behind the Design

- Common Core Standards Writing Team (2011). *Progressions for the Common Core State Standards in Mathematics: 6–7, Ratios and Proportional Relationships*. — ratio tables and scaling by non-whole factors as the path from 6.RP.A.3 toward proportional relationships.
- Lobato, J., & Ellis, A. B. (2010). *Developing Essential Understanding of Ratios, Proportions, and Proportional Reasoning for Teaching Mathematics: Grades 6–8*. NCTM. — ratios as multiplicative comparisons; the additive-thinking error.
- Lamon, S. J. (2012). *Teaching Fractions and Ratios for Understanding* (3rd ed.). Routledge. — part-to-part and part-to-whole reasoning with mixtures.
- Moschkovich, J. (2015). Academic literacy in mathematics for English learners. *Journal of Mathematical Behavior*, 40, 43–62. — the word-study and sentence-frame design of the worksheet.